

Task 1

Question 1 : location for DDR register

It exists in the memory map and it controls hardware peripherals

Question 2 : SRAM vs Flash

SRAM:

Volatile memory (data is lost as soon as power is removed).

It is used as the runtime data memory for the processor.

It stores temporary data, including variables, function parameters, and the Stack.

It provides very fast read and write access compared to other memory types.

It consists of transistor-based memory cells that do not require refreshing, unlike DRAM.

Flash Memory

Non-volatile memory (retains data when power is lost).

It is used to store the executable program code (the dedicated control program).

The CPU fetches instructions directly from Flash memory to execute them.

It is designed for high endurance, typically supporting up to 10,000 write/erase cycles.

Acts as the primary storage for the application firmware and constant data